

Unblackbox: Understanding Tradeoffs in Personalized Marketing Interventions

Marketing interventions are increasingly designed through experimentation and personalization that optimize firm objectives, yet firms face tradeoffs across outcomes and horizons. Existing personalization emphasizes *who* should receive an intervention more than *why* tradeoffs arise across customers. We propose a descriptive framework for heterogeneous tradeoffs: we extend recursive partitioning to two outcomes and build interpretable segments from standard A/B data, surfacing groups with different joint responses—including localized segments where one outcome improves while another worsens. The output supports organizational learning, redesign, and surfacing harms hidden in opaque targeting. Simulations show strong detection of localized contradictions in large, noisy panels with parsimonious trees versus a two-step benchmark that estimates each outcome separately and merges segment labels. We note implications for responsible personalization and accountability.

Keywords: personalized marketing, heterogeneous treatment effects, A/B testing

Intended track: Marketing Analytics and Research Method

1. Introduction

Modern firms routinely experiment with and re-optimize marketing interventions using controlled tests and live performance data. They design personalized policies—rules mapping customer characteristics to treatment—estimated to maximize scalar objectives such as conversion, revenue, or engagement (Simester et al., 2020; Lemmens et al., 2025; Ascarza, 2018). Yet the same organizations almost always care about several outcomes at once, for example lifting sales without damaging churn, satisfaction, or brand equity (Chandy et al., 2021; Hohnhold et al., 2015; Anderson and Simester, 2004).

A common operational response is to monitor *guardrail* metrics: after a policy is trained on a primary objective, analysts compare aggregate treatment effects on secondary outcomes to ensure that short-run gains are not accompanied by unacceptable deterioration elsewhere. These checks are valuable, but they evaluate a policy still designed mainly on one target; they screen population averages rather than constructing representations that use both outcomes jointly when describing heterogeneity. Another approach compresses multiple outcomes into a composite or weighted score; in practice it is often difficult for managers to commit to fixed weights when the managerial question is fundamentally about trade-offs rather than scalarization (Rafieian et al., 2025). Recent multi-objective personalization work estimates flexible heterogeneous effects and summarizes feasible policies, including Pareto frontiers, so firms can choose allocations in the language of trade-offs (Rafieian et al., 2025).

These methods are largely *prescriptive*: they emphasize *who* should receive an intervention more than *why* trade-offs arise across customers. Combined with opaque high-flexibility targeting models, that emphasis creates practical limitations. It can restrict *organizational learning*, because optimization artifacts are hard to interrogate even when they improve lift. Aggregate reporting can *hide localized harms*: small segments where a primary metric improves while a secondary worsens may be invisible in pooled estimates yet consequential for churn, complaints, or scrutiny (Lambrecht and Tucker, 2019; Helberger et al., 2020). Fully automated targeting also collapses rich strategy into a single objective, even though firms often want segment-specific balances between horizons or stakeholders. Finally, treating the intervention as fixed and focusing only on allocation sidesteps the reality that many trade-offs stem from product or creative design misalignment—understanding *where* tensions arise is often a prerequisite to redesign toward win-win improvements.

This paper takes a *descriptive* approach. Using standard randomized A/B data (X, T, Y^A, Y^B) , we build a transparent, interpretable segmentation: a small set of explicit rules that show how two outcomes co-move with the intervention—where trade-offs are large, small, or aligned. The goal is to support learning from experiments, deliberative segment-level decisions, documentation of vulnerable patterns, and intervention redesign, not only reassignment of customers.

Methodologically, we extend recursive partitioning to two outcomes (Athey and Imbens, 2016; Wager and Athey, 2018; Athey et al., 2019). The estimator builds a small set of explicit covariate rules such that segments differ jointly on estimated primary and secondary treatment

effects, with optional extra weight on the practical pattern where the primary lifts while the secondary is flat or worsens—the tension managers most often worry about under short-run optimization. The representation stays parsimonious relative to methods that fit a flexible model for each outcome separately and only merge labels afterward.

We benchmark against a two-step analogue of common practice: estimate heterogeneous effects for each outcome, derive segment labels from the two surfaces, and merge them with a classifier (Rafieian et al., 2025). Simulations show substantially better detection of that focal tension pattern, with far fewer leaves, when contradictions are rare, noise is high, and customer descriptors resemble a single high-cardinality vendor taxonomy. We contribute (i) a descriptive complement to multi-objective personalization (Rafieian et al., 2025; Huang and Ascarza, 2024), (ii) audit-relevant interpretability (Chandy et al., 2021; Lambrecht and Tucker, 2019; Raji et al., 2020), and (iii) joint partitioning that foregrounds diagnostically relevant co-movement (Künzel et al., 2019; Nie and Wager, 2021). Outputs can also support emerging accountability documentation requirements (European Union, 2022).

Section 2 reviews related literature. Section 3 presents the framework. Section 4 reports the factorial study. Section 5 discusses implications and limitations.

2. Related Literature

2.1 Multi-objective personalization and horizons

A growing literature studies interventions under competing objectives (Rafieian et al., 2025). Short-run lift metrics are easy to optimize in experiments, while longer-run outcomes are noisier and slower to observe; misalignment can reverse apparent gains when heterogeneity is ignored (Huang and Ascarza, 2024). Evidence documents segment-level contradictions that disappear in aggregates (Anderson and Simester, 2004; Hohnhold et al., 2015). Our contribution is diagnostic: we map *where* tradeoffs concentrate instead of estimating an optimal allocation.

2.2 Accountability and audits

Algorithmic systems can produce unintended distributional consequences even absent explicit demographic targeting (Lambrecht and Tucker, 2019). Regulators increasingly require risk assessment and documentation (European Union, 2022); internal auditing frameworks stress ongoing monitoring (Raji et al., 2020). Interpretable segmentations on experimental evidence can complement opaque prescriptive targeting by making localized tensions discussable.

2.3 Heterogeneous treatment effects

Causal trees, causal forests, and generalized random forests estimate effect heterogeneity for a single outcome (Athey and Imbens, 2016; Wager and Athey, 2018; Athey et al., 2019). When analysts have two outcomes, a common workflow estimates each surface separately and merges estimates downstream. That modular design can miss small contradiction segments when each single-outcome signal is too weak to pass model selection. Joint construction reallocates the partitioning objective toward the diagnostic pattern before pruning.

3. Diagnostic Framework

3.1 Problem setup

Consider a randomized experiment with binary treatment $T_i \in \{0, 1\}$, pre-treatment covariates X_i , and two outcomes Y_i^A (primary) and Y_i^B (secondary, possibly partially observed). Under randomization, within any covariate-defined segment, simple difference-in-means estimators recover conditional average treatment effects $\tau^A(x)$ and $\tau^B(x)$. The sign pattern (τ^A, τ^B) partitions the population into four qualitative regions; Region 2 ($\tau^A > 0, \tau^B \leq 0$) is the focal *contradiction* when the firm optimizes the primary but must surface harm on the secondary. The framework extends to more regions of interest and to more than two outcomes in principle; we focus on Region 2 for concreteness.

3.2 Recursive partitioning

The procedure grows a binary tree by splitting segments on single covariate thresholds. At each candidate split, primary and secondary treatment effects in each child segment are estimated by difference in means; the chosen split is the one that most increases joint separation on both margins, with optional extra weight on Region 2 (defined above) when analysts want splits to favor that harm-with-lift pattern. Effects are scaled using root-level references so the two outcomes are comparable. After a maximal tree is grown, bottom-up pruning removes weak partitions.

3.3 Practical properties

The tree is invariant to monotone transformations of covariates, accommodates mixed types in the usual partitioning sense, and performs implicit variable selection because uninformative features never improve the splitting objective. When Y^B is partially observed, $\hat{\tau}^B$ is computed on the observed subpopulation inside each candidate segment, mirroring common retention or satisfaction measurement designs.

4. Simulation Study

4.1 Design and data generating process

We use a region-stratified binary DGP with fixed factorial grids (independent seeds). Covariates are $k \in \{1, 3, 6, 10\}$ categorical variables with one-hot dimension $p=30$ (e.g., $k=1$ implies 30 levels on one variable; $k=10$ implies ten variables with three levels each). *Rareness* sets both the target observation share and combination-type share for Region 2; remaining combination mass is split equally across other regions. Large Cartesian products use a capped pool of combinations for allocation. Treatment effects follow deterministic ± 1 maps by region (intensity 1.0); i.i.d. Gaussian noise with common standard deviation σ is added to each observed outcome. Factors include $\sigma \in \{0.01, 0.1, 1, 10\}$, six rareness values from 0.01 to 0.32, sample sizes $N \in \{10,000, \dots, 320,000\}$, and ten replicates per crossed cell ($4 \times 4 \times 6 \times 6 \times 10$ design).

Each replicate is split 60%/20%/20% for training, validation, and testing. We fit Divergence-Tree for $\lambda \in \{0, 1, 2, 4, 8\}$ with Region 2 activation when $\lambda > 0$, and two-step benchmarks

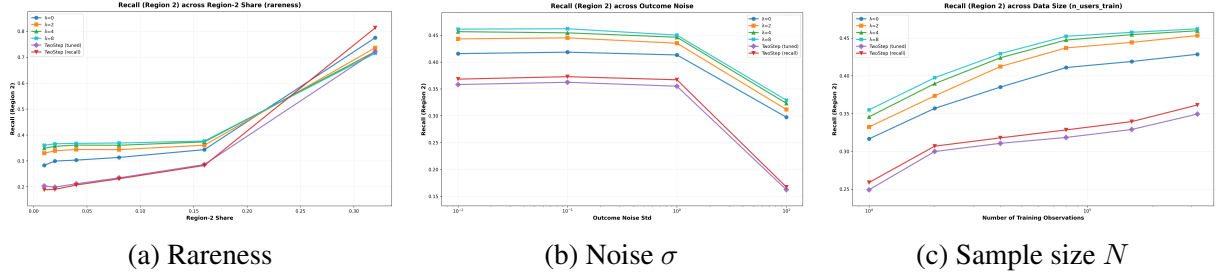


Figure 1: Region 2 recall versus design factors (means pooling other factors). DivergenceTree $\lambda=8$ vs. two-step (tuned).

(separate forests, sign regions, classification tree) tuned for accuracy and for Region 2 recall. Region labels are known only on the test split for evaluation.

4.2 Aggregate performance

Table 1 reports means over completed simulation rows. Every λ configuration dominates both two-step variants on mean Region 2 recall while maintaining higher mean accuracy; trees are far more parsimonious (about 40 versus about 120–140 leaves).

Method	Accuracy	Recall (Region 2)	FNR (Region 2)	N Leaves
DivTree $\lambda = 0$	0.522 (0.267)	0.386 (0.414)	0.614 (0.414)	41.03 (31.42)
DivTree $\lambda = 1$	0.520 (0.268)	0.401 (0.401)	0.599 (0.401)	41.07 (31.32)
DivTree $\lambda = 2$	0.517 (0.269)	0.409 (0.393)	0.591 (0.393)	40.64 (31.01)
DivTree $\lambda = 4$	0.515 (0.270)	0.420 (0.383)	0.580 (0.383)	40.72 (30.74)
DivTree $\lambda = 8$	0.512 (0.271)	0.426 (0.379)	0.574 (0.379)	40.35 (30.18)
TwoStep (tuned)	0.495 (0.153)	0.309 (0.270)	0.691 (0.270)	143.91 (86.98)
TwoStep (recall)	0.472 (0.152)	0.319 (0.297)	0.681 (0.297)	123.52 (89.91)

Table 1: Method comparison across simulation runs ($n = 5758$). Entries are mean (standard deviation). Recall for Region 2 is the primary diagnostic metric. Accuracy is reported to show the trade-off as λ increases.

4.3 Heterogeneity, λ , and vendor taxonomies

Cell-averaged Region 2 recall gaps favor $\lambda=8$ versus the accuracy-tuned two-step benchmark when Region 2 is rare and when noise is high; the gap narrows when Region 2 is abundant. Along rareness slices, $\lambda=0$ can outperform large λ when contradictions are easy to sample, matching the interpretation of λ as a scarcity-sensitive emphasis knob. At $k=1$, DivergenceTree recall averages about 0.95–0.96 across λ whereas two-step Region 2 recall averages about 0.16 with very low Region 2 F_1 , illustrating a failure mode of estimate-then-merge workflows under single high-cardinality segmentations common with third-party data.

Table 2 reports bin means underlying the plots. Figure 1 shows recall versus rareness, noise, and N for $\lambda=8$ versus two-step (tuned). Figure 2 summarizes overall method comparison for Region 2 recall.

<i>Rareness</i> (Region 2 share); recall for Region 2			
Rareness	DivTree $\lambda=8$	TwoStep (tuned)	Gap
0.01	0.360	0.203	0.157
0.02	0.366	0.199	0.167
0.04	0.368	0.212	0.156
0.08	0.369	0.235	0.134
0.16	0.377	0.286	0.091
0.32	0.715	0.722	-0.007
<i>Outcome noise σ</i> ; recall for Region 2			
σ	DivTree $\lambda=8$	TwoStep (tuned)	Gap
0.01	0.461	0.358	0.103
0.10	0.462	0.362	0.100
1.00	0.450	0.355	0.095
10.00	0.329	0.162	0.167
<i>Sample size N</i> (total per replicate); recall for Region 2			
N	DivTree $\lambda=8$	TwoStep (tuned)	Gap
10,000	0.355	0.249	0.106
20,000	0.398	0.300	0.098
40,000	0.429	0.311	0.119
80,000	0.452	0.319	0.134
160,000	0.458	0.329	0.129
320,000	0.462	0.349	0.113

Table 2: Region 2 recall at selected design levels (cell means over all other factors). *Gap* is DivTree minus two-step (tuned). Values match the aggregated v4 analyzer summaries plotted in Figure 1.

4.4 Sparsity across k

Figure 3 traces Region 2 recall across $k \in \{1, 3, 6, 10\}$. At $k=1$, the joint method maintains very high recall for every λ curve, whereas the two-step benchmarks remain near the bottom of the plot. As k increases, the latent combination space expands and all methods lose recall; at $k=10$ the two-step variants can partially close the gap in the plotted bin averages. This pattern underscores that the relative advantage of joint construction is largest in low-dimensional categorical representations typical of vendor segmentations, while high- k designs stress every estimator.

4.5 Co-movement emphasis across λ

Within DivergenceTree, increasing λ raises mean Region 2 recall from about 0.39 to about 0.43 in Table 1 while lowering mean accuracy slightly from about 0.52 to about 0.51, reflecting the intended accuracy–recall trade-off. Slicing the factorial summaries by rareness shows when Region 2 is scarce (0.01 share), $\lambda=8$ attains Region 2 recall about 0.36 versus about 0.28 for $\lambda=0$; when Region 2 is abundant (0.32), the ordering reverses ($\lambda=0$ about 0.78 recall versus about 0.72 for $\lambda=8$, with higher accuracy for $\lambda=0$). Analysts who expect contradictions to be rare should therefore lean toward larger λ ; analysts who expect contradictions to be widespread

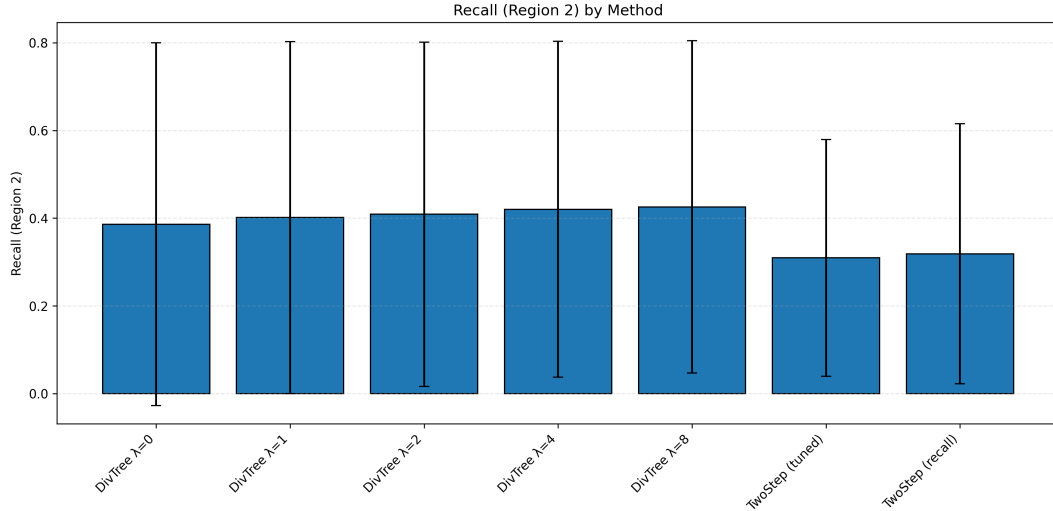


Figure 2: Overall simulation averages: Region 2 recall by method (error bars reflect Monte Carlo variability in the plot routine).

can keep $\lambda=0$. Moderate values ($\lambda \in \{2, 4\}$) retain most of the recall gain with a smaller accuracy cost in the same table.

4.6 Computational cost

The two-step method fits two tuned outcome models and a tuned label merger per replicate; wall-clock times are orders of magnitude larger on average than DivergenceTree in the logged simulation outputs, with high variance across cells. For large-scale audit grids, the joint tree is attractive not only statistically but also as a lighter repeatable screen before committing to heavier benchmarks.

5. Discussion

The framework is descriptive: it turns experimental evidence into explicit segment rules for learning and accountability, complementing prescriptive personalization (Rafieian et al., 2025). Outputs can support documentation of localized harms alongside aggregate guardrails and connect to emerging platform obligations (European Union, 2022).

The simulation mirrors a transparent factorial stress test; practitioners should view absolute performance levels as calibration-dependent. Nonetheless, the recurring ordering—joint tree ahead on Region 2 recall with fewer leaves, two-step deteriorating sharply under noise and when many levels load on a single descriptor field—is stable across the wide grid we evaluate. Reproducibility is supported by fixed seeds, archived configuration files, and aggregated result tables in the repository.

Limitations include the two-outcome focus in this submission, reliance on randomization in the baseline, and analyst choices about which tension patterns to emphasize when growing trees. Extensions include additional outcomes, observational variants with doubly robust or propensity-adjusted splitting, and coupling diagnostics to constrained re-optimization after diagnosis.

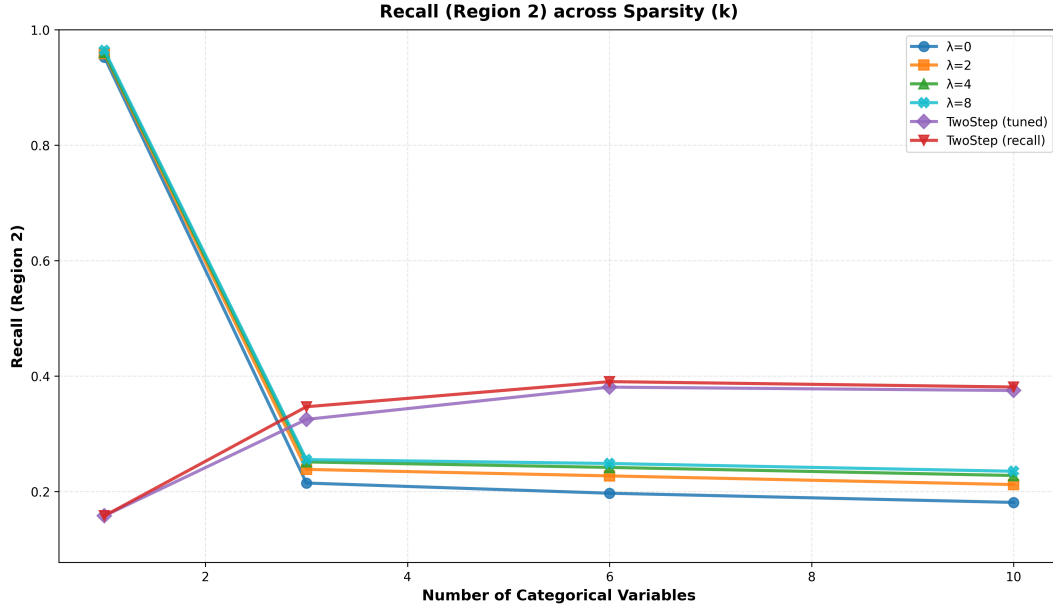


Figure 3: Region 2 recall versus sparsity k (number of categorical variables) with other factors averaged in each bin.

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